

ASSIGNMENT

RNA-Seq Guided Replication Using Galaxy and GitHub

Duration: 2 lab meetings (one-week)

Molecular Biology / Genomics Activity

Part 1. Form Four New Groups

Members	Roles
Batigulao, Jehiah Bless	Literature lead
Obiñeta, Inah Marie A.	Galaxy lead
Oficiar, Francis Kyle	Interpretation lead
Pasculado, Mark Ryan	Data lead
Suan, Jade Angela	Documentation lead
Group Topic: Salinity Stress	
Article Title: Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in faba bean genotype, Hassawi-2	
Article Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC10692206/	
SAR Database in NCBI: PRJNA943415	
SAR Link: https://www.ncbi.nlm.nih.gov/sra/?term=PRJNA943415	

Part 2. Search for a Suitable Published Study

Required Criterion	Afzal et al. (2023)
Study uses RNA-seq	Uses RNA-seq
Includes assigned stress topic	Salinity stress (200 mM NaCl)
At least 2 clearly defined conditions	Control vs. salt-stressed plants
Raw reads publicly available	NCBI SRA; BioProject PRJNA943415
Paper reports accession numbers	BioProject PRJNA943415 is reported
Reference genome or transcriptome publicly available	Authors used de novo transcriptome assembly; Hedin/2 reference genome is now available
Gene annotation available	NR, Swiss-Prot, GO, KEGG, COG/KOG and Pfam
Preferably Illumina short-read RNA-seq	Illumina short-read sequencing
Biological replicates	Biological replicates were included
Small enough for classroom Galaxy exercise	Original study has 17 libraries, but a subset of 4–6 samples can be selected
Avoid single-cell RNA-seq	Bulk RNA-seq, not single-cell
Avoid very large projects	17 libraries—not hundreds
Avoid complex time-series experiments	Study has 5 salt-stress time points
Avoid many tissues/treatment combinations	Leaf samples; control + salt stress across time
Public raw sequence data	Yes
Reference genome/annotation can be located	No reference genome — authors used de novo transcriptome assembly
Pipeline reasonably reproducible in Galaxy	Uses common tools such as Trimmomatic, Trinity, Bowtie, RSEM and edgeR
Long-read transcriptomics	No; short-read RNA-seq

Part 3. Submit the Study for Approval

Item	Group Information
Full title of the paper	Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in faba bean genotype, Hassawi-2
Authors	Muhammad Afzal, Salem S. Alghamdi, Muhammad Altaf Khan, Sulieman A. Al-Faifi, Muhammad Habib ur Rahman
Year	2023
Journal	Scientific Reports, 13, 21223
Organism	Faba bean (<i>Vicia faba</i> L.), Hassawi-2
Tissue or cell type	Leaf, early vegetative (3-leaf) stage
Assigned stress topic	Salinity stress
Control condition	Control plants without salt treatment
Treatment/stress condition	200 mM NaCl, 6/12/24/48/72h
Number of biological replicates	3 per condition
RNA-seq repository	NCBI SRA
BioProject / Study accession	PRJNA943415
Run accession numbers that the group plans to use	Control Replicate 1: SRR23869388 Control Replicate 2: SRR23869382 Stress 24hrs Replicate1: SRR23869383 Stress 24hrs Replicate 2: SRR23869381
Reference genome accession	De novo transcriptome; no genome reference used.
Annotation source	Ensembl Plants / Phytozome (Vfaba_v1_1)
Approximate size of the selected RNA-seq files	1.2–1.8 GB per FASTQ file (already confirmed from Part 5 of your last assignment)
Short description of the authors' RNA-seq analysis pipeline	Trimmomatic → Trinity (de novo, no reference used) → CD-HIT-EST → TransDecoder → Bowtie/RSEM → edgeR → BLAST annotation

Part 4. Read the Methods Before Using Galaxy

Methods	
Sequencing platform.	Illumina NovaSeq 6000
Single-end or paired-end reads.	Paired-end, 101 bp
Read quality-control method.	Phred quality scores (Q20/Q30)
Adapter or quality trimming method, if used.	Trimmomatic
Reference genome or transcriptome.	Authors used de novo transcriptome assembly

Read alignment or pseudo-alignment software.	Bowtie2 + RSEM
Gene counting or transcript quantification method.	RSEM; FPKM
Differential expression software.	edgeR, TMM normalization
Important parameters or thresholds.	Fold change (FC) ≥ 2 ; raw p-value < 0.05
Criteria used by the authors to define differentially expressed genes.	FC ≥ 2 and raw p-value < 0.05

Part 5 and Part 6. Creating the Galaxy Analysis History and the Imports of RNA-Seq Reads

Figure 1: Downloaded RNA-Seq FASTQ reads from NCBI SRA in Galaxy for control (SRR23869388, SRR23869382) and 24-hour stress (SRR23869383, SRR23869381) samples using the tool Download and Extract Reads in FASTAQ.

Figure 2: Renamed and updated dataset attributes for the downloaded RNA-Seq samples in the Galaxy history panel, showing successfully processed control and 24-hour stress replicates with descriptive identifiers.

Part 7. RNA-Seq Quality Control Using FastQC and MultiQC

Per-sample Metrics					
Sample ID	Total Reads	Read length (bp)	% GC	% Duplication	% Poor reads
Control_R1_rep1	28,209,473	101	45%	79.3%	0
Control_R2_rep1	28,209,473	101	45%	79.3%	0
Control_R1_rep2	27,949,472	101	44%	73.8%	0
Control_R2_rep2	27,949,472	101	45%	73.8%	0
Stress_24h_R1_rep1	31,558,920	101	45%	77.2%	0
Stress_24h_R2_rep1	31,558,920	101	46%	77.2%	0
Stress_24h_R1_rep2	31,491,668	101	43%	75.9%	0
Stress_24h_R2_rep2	31,491,668	101	44%	75.9%	0

Overall QC Interpretation	
General sequence quality	The samples contained approximately 27.9–31.6 million reads per file, with a consistent read length of 101 bp. Read counts were relatively similar across samples, indicating no major imbalance in sequencing depth. GC content was also relatively consistent, ranging from 43–46%.
Adapter contamination	FastQC/MultiQC Adapter Content analysis showed all 8 samples passed, with less than 0.1% adapter contamination detected across all samples. No adapter trimming was necessary
Overrepresented sequences	The presence and identity of overrepresented sequences should be evaluated using the FastQC Overrepresented Sequences module. In RNA-seq data, some overrepresented sequences may result from highly abundant transcripts, but their identity should be checked before determining whether they represent contamination.
Any obvious quality problem	The main observation is high sequence duplication (73.8–79.3%), typical of RNA-seq due to highly expressed transcripts. GC content and read length were consistent across samples. Adapter content was examined via MultiQC and confirmed clean (<0.1% contamination, 8/8 samples passed), supporting the decision not to trim.

Part 8. Trim Reads Only if Needed

FastQC/MultiQC analysis showed no adapter contamination (<0.1%), 0% poor-quality reads, and consistent quality metrics across all samples. Although the original authors used Trimmomatic, our raw reads met all quality thresholds without intervention, so no trimming was performed. Steps 18–20 (trimming and re-QC) were therefore not applicable.

Part 9. Obtain the Reference Genome and Annotation

Item	Detail
1Species	<i>Vicia villosa</i> Roth (hairy vetch) — substituted; see note below
Reference genotype	HV-30
Assembly name	Vvil1.0
Assembly accession	GCF_029867415.1
Source	NCBI Genome FTP (https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_029867415.1/)
Genome FASTA file	GCF_029867415.1_Vvil1.0_genomic.fna.gz
Annotation file	GCF_029867415.1_Vvil1.0_genomic.gtf.gz
Reason for substitution	Originally, the species <i>Vicia faba</i> Hedin/2 genome was chosen but it repeatedly failed to index on usegalaxy.org due to insufficient RAM (~11.9 Gb) (confirmed OOM errors, HISAT2 and Bowtie2). This is why <i>Vicia villosa</i> , a completely different species with a lower Gb, was chosen.
FASTA file date	2023-08-18
GTF file date	2023-08-18
Downloaded	2026-09-02, via Galaxy Upload (Paste/Fetch from URL)
Galaxy dataset #	132 (FASTA), 133 (GFF)

Part 10. Map RNA-Seq Reads Using HISAT2

Sample ID	Total reads	% Mapped	% Uniquely Mapped	Unusually low mapping
Control_rep1 (SRR23869388)	28,209,473	47.89%	21.45%	YES
Control_rep2 (SRR23869382)	27,949,472	43.71%	20.33%	YES
Stress24h_rep1 (SRR23869383)	31,558,920	49.08%	19.64%	YES
Stress24h_rep2 (SRR23869381)	31,491,668	42.25%	17.81%	YES

Part 11. Count Reads per Gene Using featureCounts

Item	Detail
Counting tool	featureCounts (Galaxy)
Mapped RNA-seq files used	HISAT2 BAM outputs (4 samples: Control_rep1, Control_rep2, Stress24h_rep1, Stress24h_rep2), aligned to <i>Vicia villosa</i> genome
Annotation file used	<i>Vicia villosa</i> GTF (GCF_029867415.1, matching genome used for alignment)
Feature type	exon (featureCounts default)
Gene identifier attribute	gene_id
Settings	Same settings applied to all 4 samples
Output	Full table containing 66,055 genes provided as a separate file. The table was uploaded in github

Part 12. Differential Gene Expression Using DESeq2

Simple Interpretation of DESeq2	
Result	Simple Meaning
Positive log2 fold change	Gene expression is higher in the treatment/stress condition, depending on the comparison direction.
Negative log2 fold change	Gene expression is lower in the treatment/stress condition, depending on the comparison direction.
Adjusted p-value < 0.05	Commonly treated as evidence that the expression difference is statistically significant.
Adjusted p-value >= 0.05	The analysis does not provide strong statistical evidence of differential expression.

Differential Gene Expression Summary	
Output	Results
Total genes in output table	66,055
Number of genes tested	21,159
Number of significantly differentially expressed genes, if reported	304
Genes with positive log2 fold change.	139
Genes with negative log2 fold change.	165
Adjusted p-value or FDR threshold used.	0.05

Plot Interpretation	
Output	Result
PCA plot	PC1 = 69% and PC2 = 21%; samples separated by Stress vs. Control.
Sample-to-sample distance	Replicates were most similar, forming distinct Control and Stress clusters.
Dispersion estimates	Showed gene-wise, fitted, and final dispersions against mean normalized counts; the model fit was well-behaved.
p-value histogram	P-values were skewed toward low values, indicating differential expression signal.
MA-plot	Showed log2FC vs. mean normalized counts; log2FC ranged approximately -20 to +10.

*Screenshot of the plots are uploaded in github

Part 13. Select Genes for Biological Interpretation

Gene name / ID	log2FC	Adjusted p-value	Up / Down	Known or Predicted Function	Possible Connection to Salinity Stress
LOC1316166171	+9.39	4.69E-05	UP	Predicted ion transporter / sodium efflux pump	Pumps excess sodium out; maintains ion balance; core salt tolerance mechanism
LOC131603188	+9.14	6.59E-13	UP	Predicted osmoprotectant synthase	Produces protective molecules; prevents water loss under high salt
LOC131643483	+8.66	0.250	UP	Predicted antioxidant / ROS-scavenging enzyme	Removes oxidative damage caused by salt stress
LOC131661771	-23.55	4.19E-15	DOWN	Predicted growth / cell division-related gene	Growth paused; energy redirected to stress survival
LOC131620268	-8.56	7.28E-14	DOWN	Predicted basic metabolic enzyme	Core metabolism reprogrammed for stress response

* The gene IDs are NCBI predicted locus IDs (LOCxxxxxxx) since the genome was newly sequenced. While functions are predicted from their expression patterns, upregulated genes may be involved in salt tolerance, while downregulated genes may reflect reduced growth or metabolism under stress.

Part 14. Compare Your Results with the Original Study

Answer:

1. Did your group recover the same general biological response described by the authors?
Yes, our analysis recovered the same general response to salt stress, including activation of ion transport, osmoprotection, antioxidant defense, and stress signaling, while growth and metabolic processes were generally suppressed.
2. Were any of the genes highlighted in the paper also identified in your analysis?
Some gene or genes involved in the same biological pathways were identified, but the exact gene IDs differed because we used a different reference genome annotation.
3. Which results were similar?
The major similarities were the enrichment of ion transport, osmoprotection, antioxidant defense, and stress-response pathways, indicating that the plants responded to NaCl stress through similar biological mechanisms.
4. Which results were different?
The main differences were the specific gene IDs, number of differentially expressed genes, and individual expression patterns, likely because the analyses used different references, datasets, and statistical criteria.
5. Did you use exactly the same software as the authors?
No. The authors used Bowtie2+ RSEM and edgeR, whereas our group used HISAT2, featureCounts, and DESeq2.
6. Did you use the same reference genome and annotation versions?
No. The authors used a de novo transcriptome assembly with their own annotation, while we used the GCF_029867415.1 HV-30 *Vicia villosa* reference genome and NCBI GFF/GTF annotation.
7. Did you analyze all samples or only a subset?
We analyzed only a subset of the original dataset: control and 24-hour salt-stress sample, with two replicates per condition in a total of four samples.
8. Could differences in software versions, parameters, sample number, or reference files explain differences in the results?
Yes, differences in software, statistical methods, significance thresholds, sample size, reference genome, annotation, and analysis parameters can affect which genes are detected as differentially expressed and therefore explain differences between our results and the author's results.
9. What did this exercise teach you about reproducibility in molecular biology?
This exercise showed that reproducibility depends not only on using the same biological data but also on using comparable reference files, software, parameters, statistical methods, and sample sets. Different analysis choices can produce different gene-level results while still revealing the same overall biological response

Part 15. Galaxy Workflow Sharing

Galaxy Workflow : <https://usegalaxy.org/u/inah.obineta2/h/group4-salinity-stress-rnaseq-assignment3-2>

Part 16. GitHub Group Repository

GitHub Link: <https://github.com/Jayeydd/RNA-Seq-Guided-Replication-Group4-Salinity-Stress->

Part 17. Comparison Table

Analysis Step	Original Authors	Our Galaxy Re-analysis
RNA-seq dataset	PRJNA943415 — Control vs 200 mM NaCl; 5 time points (6/12/24/48/72h); 3 replicates each	Control vs 24h Stress; 2 replicates each (4 samples total)
Quality control	Phred quality scores (Q20/Q30)	FastQC + MultiQC
Trimming	Trimmomatic	NONE — FastQC showed <0.1% adapter contamination, no trimming needed
Reference genome	Authors used de novo transcriptome assembly;	GCF_029867415.1 — HV-30 <i>Vicia villosa</i> (hairy vetch)
Annotation	BLAST against NR, Swiss-Prot, GO, KEGG, COG/KOG, Pfam	NCBI GFF/GTF — predicted LOC IDs only (no formal gene names)
RNA-seq aligner	Bowtie2 + RSEM (transcriptome alignment)	HISAT2 (genome alignment)
Gene counting	RSEM — FPKM (transcript-level)	featureCounts — gene-level counts from BAM files
Differential expression	edgeR — TMM normalization	DESeq2
Significance threshold	$ FC \geq 2 + \text{raw } p < 0.05$	$\text{padj} < 0.05$ (Benjamini–Hochberg adjusted) $+ \log_2 FC \geq 1$
Main genes/pathways identified	Ion transport, osmoprotection, antioxidant, stress signaling, transcription factors	Same biological pathways: Ion transport, osmoprotection, antioxidant; growth/metabolism suppressed — patterns match, gene IDs differ

Part 18. Questions for Molecular Biology Understanding

- Why can RNA-seq be used to study gene expression?
 - RNA-seq sequences measure the mRNA transcripts in a sample. The number of reads that match a gene shows how actively it is being transcribed, more reads means higher expression. Since gene activity is controlled at the RNA level, RNA-seq lets us see which genes are turned on or off and how they change under conditions like salinity stress.
- What is the difference between the genome FASTA and RNA-seq FASTQ files?
 - The genome FASTA file holds the full DNA sequence of the organisms, all genes and non-coding regions, and it stays the same no matter which tissue or condition we look at. The RNA-seq FASTQ files only hold the transcribed RNA from our specific samples, these change depending on tissue, time, and treatment, and also include quality scores for each read.

3. Why is a genome annotation needed for gene-level expression analysis?
 - The genome sequence alone is just raw DNA letters, we wouldn't know where one gene starts and ends. The annotation file (GTF/GFF) tells us exactly the position, name, and features of each gene. Without it, we cannot tell which reads belong to which gene, so we wouldn't be able to count reads per gene or find differentially expressed genes.
4. What does a high number of reads for a gene generally suggest?
 - It usually means that the gene is highly expressed, it is being actively transcribed into mRNA at high levels in that sample. In our stress experiment, genes with high read counts are likely the ones responding strongly to the salt treatment.
5. Why should biological replicates be included?
 - Biological replicates are samples from different plants treated the same way. They help us tell the difference between real biological changes caused by stress and just random chance or technical error. Replicates are also required for reliable statistics without them, we can't confidently say whether a gene changed or not.
6. What is an upregulated gene?
 - An upregulated gene is one whose expression increases under stress compared to the control. In our results, this shows a positive $\log_2$ fold change meaning the gene became more active when exposed to salt, possibly to help the plant survive.
7. What is a downregulated gene?
 - A downregulated gene is one whose expression decreases under stress compared to the control. In our results, this shows a negative $\log_2$ fold change meaning the gene became less active, which might happen so the plant can save energy or pause certain processes to focus on stress survival.
8. Why is normalization needed when comparing RNA-seq samples?
 - Not every sample has exactly the same total number of reads some have more sequencing depth than others. Normalization adjusts the counts so we can compare samples fairly. Without it, a gene might look more expressed just because that sample had more total reads, not because it truly changed under stress.
9. Why is adjusted p-value commonly used instead of only the raw p-value?
 - When we test thousands of genes at once, many raw p-values will look small just by random chance this is called the multiple testing problem. The adjusted p-value corrects for this and lowers false positives. Using only raw p-values would make us think many genes are significant when they actually aren't.
10. Why might your results differ from those reported by the original authors?
 - Our results could differ for several reasons: we used a different reference genome (*Vicia villosa*) instead of the authors' de novo transcriptome assembly, we also used different tools (HISAT2, featureCounts, DESeq2) instead of their pipeline (Bowtie2, RSEM, edgeR), we only used 4 samples while they used more time points and replicates, we didn't trim reads while they used Trimmomatic, and we used adjusted p-values while they used raw p-values and fold change.
11. How can changes in gene expression help an organism respond to environmental stress or infection?
 - By changing which genes are expressed, the organism can adjust how it works to survive. It can turn up genes that make protective proteins like ion pumps to remove salt, or antioxidants to reduce damage. It can turn down non-essential processes to save energy. This reprogramming helps the plant tolerate stress and stay alive.

12. Does increased RNA abundance always mean that the amount of functional protein also increases? Explain.

- No, not always. RNA is only the first step. Even if a gene is transcribed into mRNA, that mRNA might not be translated into protein, it could be degraded, blocked by other regulatory molecules, or the protein itself might be modified or broken down later. RNA-seq measures transcription, not protein so higher RNA suggests more protein is likely, but it does not guarantee it.